



A modal decomposition and expansion approach for prediction of dynamic responses on a monopile offshore wind turbine using a limited number of vibration sensors



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ARTICLE INFO

Article history:

Received 10 November 2014

Received in revised form

26 May 2015

Accepted 20 July 2015

Available online 14 August 2015

Keywords:

Modal decomposition

Modal expansion

Response estimation

Structural health Monitoring

Offshore wind turbines

Foundations

ABSTRACT

Structural health monitoring of wind turbines is usually performed by collecting real-time operating data on a limited number of accessible locations using traditional sensors such as accelerometers and strain-gauges. When dealing with offshore wind turbines (OWT) though, most of the fatigue sensitive spots are inaccessible for direct measurements, e.g. at the mudline below the water level. Response estimation techniques can then be used to estimate the response at unmeasured locations from a limited set of response measurements and a Finite Element Model. In this paper the method will be validated using accelerations only. The method makes use of a modal decomposition and expansion algorithm that allows for successful response prediction. The algorithm is first validated using simulated datasets provided from HAWC2 and then using real time data obtained from a monitoring campaign on an offshore Vestas V90 3 MW wind turbine on a monopile foundation in the Belgian North Sea.

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1. Introduction

Online monitoring of wind turbines is a more and more critical issue as the machines are growing in size and offshore installations are becoming more common. To increase power generation and limit the weight, the turbines are becoming structurally more flexible, thus an accurate prediction of their dynamic behavior is mandatory.

Offshore wind turbines are exposed to a variety of dynamic loading scenarios for which fatigue accumulation needs to be monitored and tracked over the course of the life of the structure [1]. Generally, the loadings are of a level and occurrence to be considered normal routine loads and the design of the structure is built to accommodate these conditions [2]. Several contemporary aeroelastic simulation tools coupled with structural dynamics models enable designers to detect, understand and solve most of the possible problems at early stages and optimize their designs [3–12]. However, there are often other environmental and operational conditions (EOC) which are more severe and more important to the overall load and fatigue accumulation for the structure. Considering the fact that the complicated interactions among different parts of the structure

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are represented better through real response measurements, several tests in laboratory scale have been conducted as an attempt to assess the integrity and the dynamic behavior of the structure [13–17]. In all the aforementioned surveys, the excitation is exactly known or/and the responses can be measured in a sufficient large number of points.

Unlike small scale applications, in the case of full scale monopile offshore wind turbines, fatigue sensitive spots are located in sections where mounting sensors are impossible or practically unfeasible (e.g. the mudline 30 m below the water level). Thus, an important issue when performing continuous monitoring of an offshore wind turbine is the limited availability of operational measurement data due to the limited set of physical sensors distributed over the turbine components. The limited sets of sensors (accelerometers and strain gauges) are insufficient to accurately depict the true stress–strain imposed on the structure. In order to assess the actual dynamic behavior and response of the structure, more sophisticated utilization of the limited data available needs to be performed. The issue of limited information due to limited availability of operational data is overcome by using an updated and properly calibrated finite element model. The calibration is performed by comparing the experimentally obtained mode shapes with the corresponding numerical mode shapes in terms of the Modal Assurance Criteria (MAC) [18,19]. As long as the finite element model is calibrated, the combined use of operational acceleration data and mode shape components derived from the finite element model is able to provide sufficient information for the prediction of accelerations at different levels along the height of the structure [20]. The prediction is based upon a modal decomposition of the measured accelerations that results in the estimation of the modal coordinates [21]. The relation between the modal coordinate and the acceleration in an arbitrary point is established by making use of the numerically obtained mode shapes [21]. This paper will demonstrate and validate the proposed approach for the continuous monitoring of the vibration levels of offshore wind turbines using both virtual and real data. First, different virtual wind-wave loading conditions with their resulting responses of the wind turbine, obtained from the fully coupled hydro-aero-servo-elastic simulation tool HAWC2, will be used for validation. Then, different real-time operating cases of the wind turbine will be examined, such as rotating, parked or idling conditions as well as an overspeed test of the rotor. All experimental data has been obtained during a long-term monitoring campaign on an offshore wind turbine in the Belgian North Sea. State-of-the art operational modal analysis techniques and the use of appropriate vibration measurement equipment allowed obtaining high quality acceleration data and accurate estimates of the natural frequencies, damping ratios and mode shapes. The paper is structured as follows. First, the necessary theory followed by the main features of the tested turbine and utilized measurement systems are described. Then the identified modal parameters using state of the art Operational Modal Analysis (OMA) are presented. Next, the finite element model is described. It will be shown that the finite element model is in good agreement with the experimental model in terms of modal characteristics. Finally, the response prediction theory will be both validated by HAWC2 simulations and using the real measurements.

2. Theory

2.1. Modal decomposition

If we consider a model of a structure with no damping, the equation of motion of the structure subjected to a force $f(t)$ is given by

$$M\ddot{\mathbf{x}}(t) + K\mathbf{x}(t) = \mathbf{f}(t) \quad (1)$$

The eigenvalue equation is then expressed as follows:

$$M\boldsymbol{\varphi}_i\omega_i^2 = K\boldsymbol{\varphi}_i \quad (2)$$

where M is the mass matrix, K is the stiffness matrix, ω_i is the eigenvalues, $\boldsymbol{\varphi}_i$ is the mode shape vectors and $\mathbf{a}(t)$, $\mathbf{x}(t)$, $\mathbf{f}(t)$ are the acceleration, displacement and force vectors respectively.

By making use of the modal decomposition approach [22–24], the displacement vector can be written as a linear combination of the mode shape vectors

$$\mathbf{x}(t) = \sum_{i=1}^n \boldsymbol{\varphi}_i q_i(t) \quad (3)$$

or in matrix form as follows:

$$\begin{aligned} \mathbf{x}(t) &= \Phi \mathbf{q}(t) \\ \Phi &= [\boldsymbol{\varphi}_1 \quad \boldsymbol{\varphi}_2 \quad \dots \quad \boldsymbol{\varphi}_n] \\ \mathbf{q}(t) &= [q_1(t) \quad q_2(t) \quad \dots \quad q_n(t)]^T \end{aligned} \quad (4)$$

where $\mathbf{q}(t)$ is the vector of the displacement modal coordinates for each time instance t , Φ is the mode shape matrix, n is the total number of modes and $[\cdot]^T$ denotes the transpose of a matrix. For most practical problems though, it is not necessary to use all mode shapes, since a good approximate solution can be obtained via modal superposition with only few mode shapes. In the current paper only a subset of modes is taken into account. In particular our interest lies within the lower frequency range and thus the first two or three modes in the Fore–Aft (FA) or Side–Side (SS) direction are considered. The aforementioned can be expressed as follows:

$$\mathbf{x}_{FA}(t) = \Phi_{FA} \mathbf{q}_{FA}(t) \quad (5)$$

$$\mathbf{x}_{ss}(t) = \Phi_{ss} \mathbf{q}_{ss}(t) \quad (6)$$

The modal coordinates contain information about the modes that are dominantly contributing to the total vibrational response and can provide insight on how many and which specific modes are contributing in certain frequency bands.

2.2. Modal expansion

If we consider a set of n_{dof} finite element degrees of freedom (DOF's) composed of a subset of a active DOF's corresponding to the measured locations of the real experimental model and d remaining DOF's corresponding to predicted locations, the displacement mode shape matrix of the related finite element model (FEM) can be written as follows:

$$[\Phi]_{FEM} = \begin{bmatrix} [\Phi_m] \\ [\Phi_p] \end{bmatrix}_{FEM} \quad (7)$$

where subscript m corresponds to the measured DOF's and p to the predicted DOF's. $\Phi_m \in \mathbb{R}^{a \times n_m}$, $\Phi_p \in \mathbb{R}^{d \times n_m}$ are the displacement mode shapes of the n_m modes considered in the analysis at the active and deleted degrees of freedom respectively. The total number of DOF's n_{dof} is equal to the sum of a active DOF's and d deleted or remaining DOF's.

Taking into account the Modal Assurance Criterion (MAC), if there exists a high correlation between the numerically obtained mode shapes and the experimentally obtained mode shapes at the measured degrees of freedom ($[\Phi_m]_{FEM}$ and $[\Phi_m]_{exp}$), the use of the modal parameters of the calibrated finite element model is allowed for any DOF of the structure. Hereafter, the subscript FEM when considering mode shapes will be omitted unless is differently clarified in the text.

The relationship for only the a DOF's in the measured locations is then written as

$$\mathbf{x}_m(t) = \Phi_m \mathbf{q}(t) \quad (8)$$

Assuming that the number of modes, n_m , is less than then number of active DOF's, a , the modal coordinates are calculated by using the pseudo inverse as follows:

$$\mathbf{q}(t) = (\Phi_m^T \Phi_m)^{-1} \Phi_m^T \mathbf{x}_m(t) = \Phi_m^\dagger \mathbf{x}_m(t) \quad (9)$$

This result can be expanded to the predicted locations with the following expression:

$$\mathbf{x}_p(t) = \Phi_p \mathbf{q}(t) = \Phi_p \Phi_m^\dagger \mathbf{x}_m(t) \quad (10)$$

2.3. Prediction of vibrational accelerations

The prediction of vibrational accelerations is based on the same principle of modal decomposition and expansion presented in Section 2.2. Similarly to Eq. (8), the expression for the limited number of experimentally known acceleration time signals at measured locations $\mathbf{a}_m(t)$ is written as follows:

$$\mathbf{a}_m(t) = \Phi_m \mathbf{q}_a(t) \quad (11)$$

where $\mathbf{q}_a(t)$ is the vector of the acceleration modal coordinates for each time instance t . The modal accelerations $\mathbf{q}_a(t)$ are calculated by using the pseudo inverse similarly to Eq. (9) with the following expression:

$$\mathbf{q}_a(t) = (\Phi_m^T \Phi_m)^{-1} \Phi_m^T \mathbf{a}_m(t) = \Phi_m^\dagger \mathbf{a}_m(t) \quad (12)$$

Finally, the result can be expanded to the predicted locations similarly to Eq. (10):

$$\mathbf{a}_p(t) = \Phi_p \mathbf{q}_a(t) = \Phi_p \Phi_m^\dagger \mathbf{a}_m(t) \quad (13)$$

2.4. Quality indicators

The proposed modal decomposition and expansion approach are validated by using the upper sensor levels to predict the accelerations at the lowest sensor level. As such, this allows us to compare measured results with predictions. The quality indicators utilized to compare the results of the predicted and the measured signals are discussed here. First is the Time Response Assurance Criterion (TRAC) [25] which is the correlation for one DOF over all time of the predicted acceleration time domain signal ($\mathbf{a}_p^{pred}(t)$) with the measured acceleration time domain signal ($\mathbf{a}_p^{meas}(t)$).

$$TRAC = \frac{[\mathbf{a}_p^{meas}(t)^T \mathbf{a}_p^{pred}(t)]^2}{[\mathbf{a}_p^{meas}(t)^T \mathbf{a}_p^{meas}(t)][\mathbf{a}_p^{pred}(t)^T \mathbf{a}_p^{pred}(t)]} \quad (14)$$

Additionally, the frequency analog to the Time Response Assurance Criterion called the Frequency Response Assurance Criterion (FRAC) is introduced. This is the correlation for one DOF over all frequencies of the predicted complex frequency

domain signal ($\mathbf{a}_p^{\text{pred}}(f)$) with the measured complex frequency domain signal ($\mathbf{a}_p^{\text{meas}}(f)$).

$$\text{FRAC} = \frac{[\mathbf{a}_p^{\text{meas}}(f)^H \mathbf{a}_p^{\text{pred}}(f)]^2}{[\mathbf{a}_p^{\text{meas}}(f)^H \mathbf{a}_p^{\text{meas}}(f)] [\mathbf{a}_p^{\text{pred}}(f)^H \mathbf{a}_p^{\text{pred}}(f)]} \quad (15)$$

where $(\cdot)^H$ denotes the Hermitian conjugate and $(\cdot)^T$ denotes the transpose of the vector.

It will be shown that the FRAC value coincides with the TRAC value when calculated in the whole frequency range. However the added value of the FRAC index is that it can quantify the quality of the frequency domain prediction at different frequency bands of interest.

Finally, the Mean Absolute Error over n_s data samples in the time domain (MAE_{TD}) and over n_f frequency samples in the frequency domain (MAE_{FD}) are used and defined according to the following formulas:

$$MAE_{TD} = \frac{1}{n_s} \sum_t |\mathbf{a}_p^{\text{meas}}(t) - \mathbf{a}_p^{\text{pred}}(t)| \quad (16)$$

$$MAE_{FD} = \frac{1}{n_f} \sum_f |\mathbf{a}_p^{\text{meas}}(f) - \mathbf{a}_p^{\text{pred}}(f)| \quad (17)$$

2.5. Overview of the approach

The monitoring process cycle of the offshore wind turbine includes 4 steps:

- Collection of continuous acceleration data from the OWT.
- Identification of modal parameters of the OWT.
- Tuning of a finite element model of the OWT.
- Response prediction at different locations on the OWT.

In order to evaluate the efficiency of the proposed modal decomposition and expansion approach for the prediction of dynamic responses of a monopile wind turbine, it was decided first to perform some virtual test simulations with HAWC2 aeroelastic code. HAWC2 (Horizontal Axis Wind turbine simulation Code 2nd generation) is an aeroelastic code developed by the Aeroelastic Design Research Program at DTU Wind Energy, DTU Risø Campus in Denmark. This code is intended for calculating wind turbine response in time domain when the wind turbine is subjected to aerodynamic, hydrodynamic and soil loads. In the current work, it is utilized as a tool for performing simulations of the examined Vestas V90 3 MW wind turbine of the Belwind farm. For more information about the whole model and how it was tuned to match the examined wind turbine, the reader is referred to [26]. Once the approach is fully validated with the virtual test data, it is further exploited for the measured datasets that correspond to real-life loading conditions of the wind turbine. All the aforementioned steps are discussed in detail in the sections that follow.

3. Offshore measurements

The measurement campaign is performed at the Belwind wind farm, which consists of 55 Vestas V90 3 MW wind turbines. The wind farm is located in the North Sea on the Bligh Bank, 46 km off the Belgian coast, Fig. 1.

The hub-height of the wind turbine is on average 72 m above sea level. Each transition piece has a height of 25 m and a weight of 120 ton. The tests are performed on the BBC01-turbine that is located in the north of the wind farm directly next to the offshore high voltage substation (OHVS). The wind turbine is placed on a monopile foundation structure with a diameter of 5 m and a wall thickness of 7 cm. The actual water depth at the location of BBC01 is 24.03 m and the monopile has a penetration depth of 20.97 m. The soil is considered stiff and mainly consists of sand.

The structures instrumented in this campaign are the tower and transition piece. Measurements are taken at 4 levels on 9 locations using a total of 10 sensors (Fig. 1-left). The locations are chosen based on the convenience of sensor mounting, such as the vicinity of platforms. The chosen levels are 69 m, 41 m, 27 m and 19 m above Lowest Astronomical Tide Level (LAT). The interface level between the transition piece and the wind turbine is 19 m above LAT. There are six accelerometers mounted at the lower three levels (two accelerometers per level) and four accelerometers at the top-level. The chosen configuration is primarily aimed at identification of tower bending modes. The two extra sensors on the top-level are placed to capture the tower torsion. MEMS accelerometers have been selected since they have a high sensitivity and are able to measure very low-frequent signals. This is necessary considering that the modal frequencies of interest, for the wind turbine structure, are expected to be around 0.35 Hz, and the expected vibration magnitude is very low, especially during ambient excitation.

The data acquisition software allows for the continuous monitoring of the accelerations. The data-acquisition system was programmed to acquire data with a sampling ratio of 5 kHz. Considering the frequency band of interest and in order to reduce the amount of data the recorded time series have been filtered with a band-pass filter between 0.05 Hz and 5 Hz and

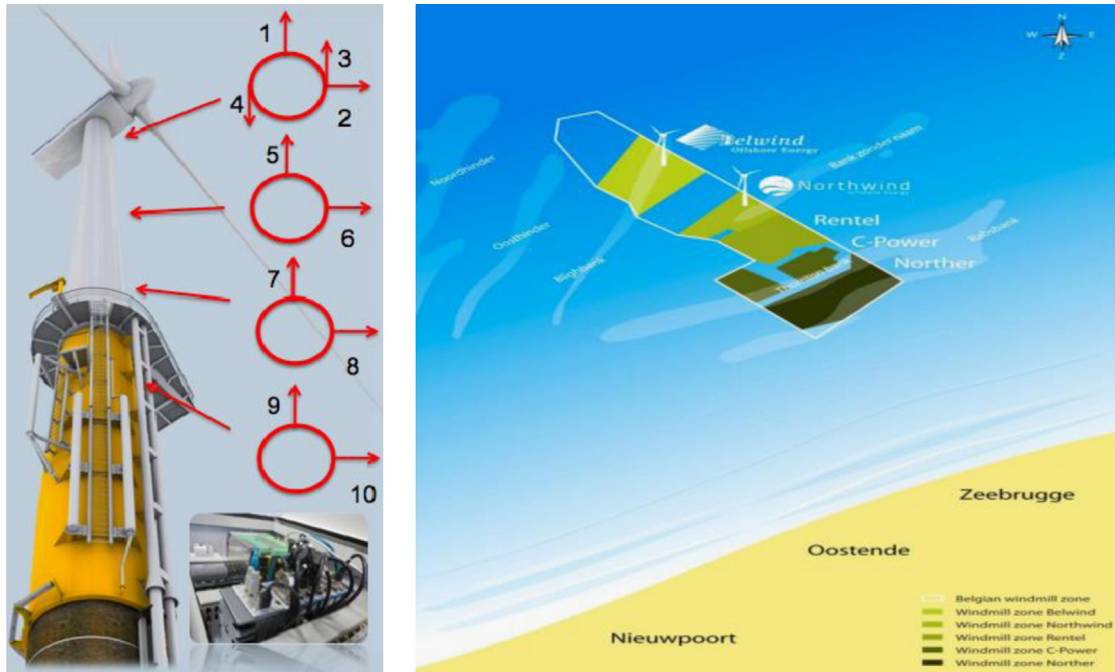


Fig. 1. A render of BBC01 offshore wind turbine at Belwind farm with indication of the measurement locations (left), location Belwind wind farm (right).

re-sampled with a sampling frequency of 20 Hz. The vibrations in the nacelle frame of reference, FA and SS, are calculated by using the yaw angle provided by the SCADA data [27]. The software measures continuously and sends data every 10 min to the server that is installed onshore using a dedicated fiber that is running over the seabed. Finally, in order to classify the operating conditions of the wind turbine during the measurements [28], both the SCADA data (power, rotor speed, pitch angle, and nacelle direction) and the ambient data (wind speed, wind direction, significant wave height, and air temperature) are being collected at 10-min intervals. Fig. 2 gives an example of the measured acceleration time domain signals in respectively rotating and parked conditions at the four levels in the Fore–Aft (FA) direction.

4. Identification of modal parameters

In the present work, a state of the art operational modal analysis technique [29–31] called pLSCF, that has been automated [32–37,39] was used to identify the modal parameters (natural frequencies, modeshapes and damping ratios) from 2 weeks of operational data while the wind turbine was in parked conditions. During these two weeks, the frequencies and mode shapes can vary due to changing environmental conditions [32,34–37]. Fig. 3 shows the mean displacement mode shapes of the 6 fundamental tower/foundation modes with their corresponding mean frequencies. The mode shapes and frequencies can be used to tune the finite element model that will be presented in the next section.

5. Finite element model modal analysis

At this stage, a numerical model that represents the real wind tower/foundation system of the examined monopile offshore wind turbine can be evaluated. A 3-D model has been set up using the commercial Finite Element Analysis software ANSYS. Fig. 4 shows the 3-D representation of the structure built up as a sequence of pipe elements. “Pipe 288” elements are used in order to take advantage of the special features allowed by this type of element (e.g. hydrodynamic loading, hydraulic added mass, and nonlinear material models). Pipe elements are geometrically defined by three parameters: height, diameter and thickness. In order to account for conical shapes, a sequence of pipes with a ‘step by step’ diminishing cross-section is used. Two assumptions have been made in order to compose a simplified model of the tower-foundation system of the wind turbine. First, the extra masses of the nacelle, rotor hub and blades are considered as point masses located at the tower top on the axis of the height of the wind turbine. Additionally, the complex interaction part that consists of steel (monopile (MP), and transition piece (TP)) as well as high resistance grout that fills the gap between the smaller diameter monopile and larger diameter transition piece is represented as a simplified element with material properties proportional to the amount of the present volume of steel and grout. The density of this simplified element called MPGRTP is calculated by considering the volumes and the corresponding weights of the individual materials present in the element (Monopile, Transition piece and grout). The other material properties (E-modulus and Poisson ratio) are tuned by assigning larger

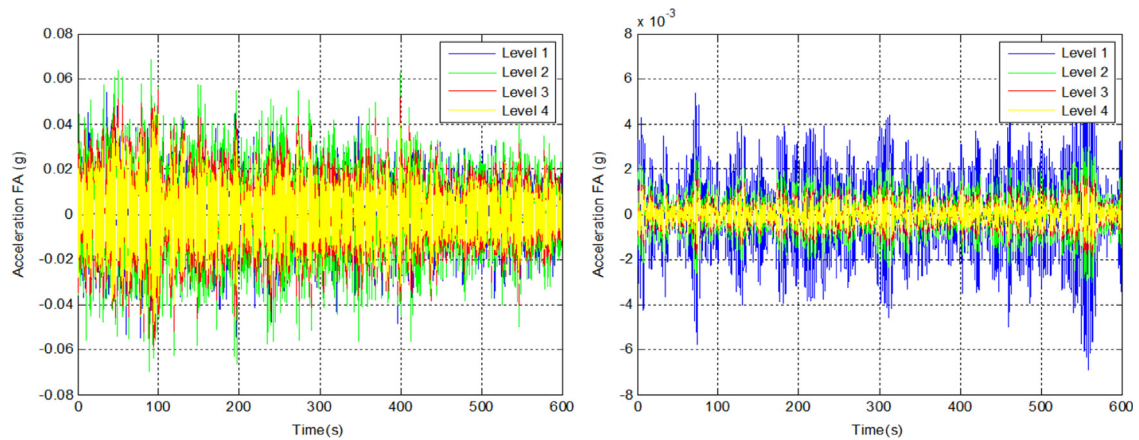


Fig. 2. Example of time domain acceleration signals in FA direction for the 4 measured levels in rotating (left) and parked (right) conditions of the wind turbine.

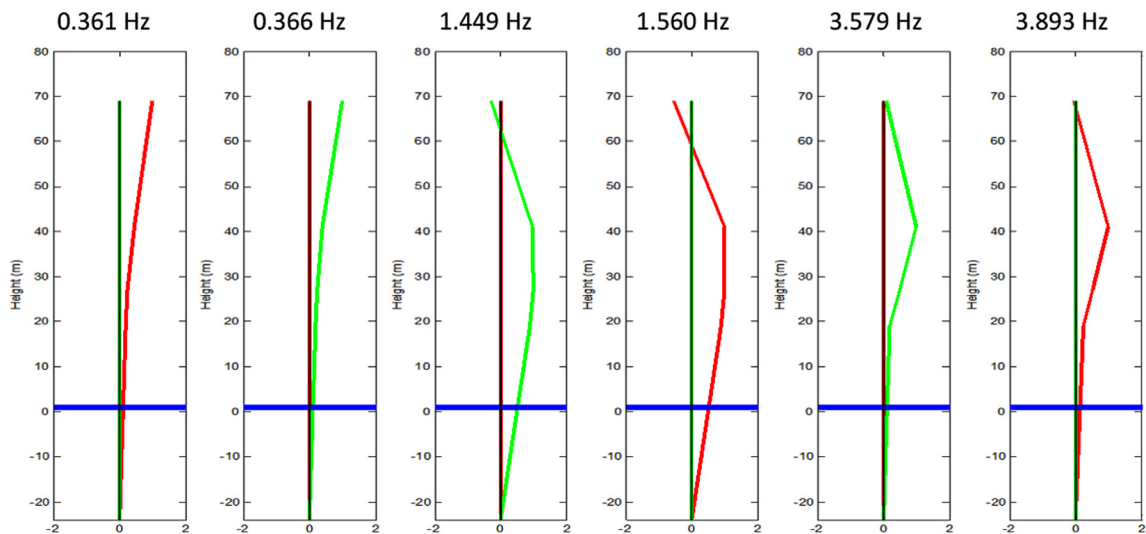


Fig. 3. Fundamental tower/foundation modes with their corresponding frequencies obtained through operational modal analysis: with red lines are depicted the modes in the FA-direction, with green lines the modes in the SS direction, blue line indicates the water level and the lowest point corresponds to the mudline level. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

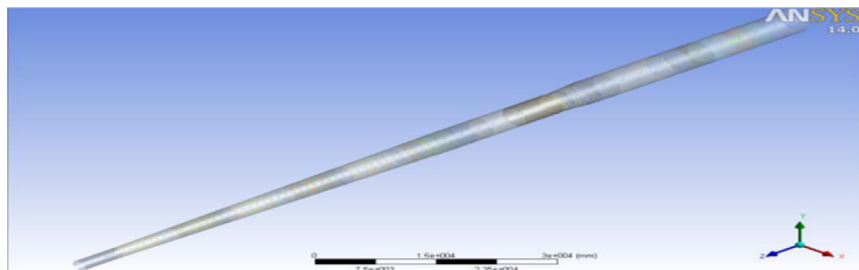


Fig. 4. Tower-foundation model of the wind turbine built up as a sequence of pipe elements.

influence on the characteristics of the steel. Structural steel has been defined with Young's modulus of 210 GPa, Poisson's ratio of 0.3 and density of 7850 kg/m³ and the grout is a cement like substance, called Ducorit S5 with Young's modulus of 60 GPa, Poisson's ratio of 0.19 and density of 2440 kg/m³. The specific properties of Ducorit S5 are obtained by adding aggregates such as quartz sand and bauxite. The weighted average of the different materials in the MPGRTP element as well as the material properties of this new element are summarized in Table 1.

The monopile foundation is driven into the sea-bed and thus it experiences a certain stiffness constraint by the surrounding soil. This soil–pile interaction is modeled by use of the Distributed spring (DS) approach [38]. The soil material properties are specified on

a layer by layer basis (Fig. 5a). Making use of these soil properties, the non-linear lateral force–displacement curve (P – Y curve) as well as the nonlinear skin friction force–displacement curve (T – Z curve) is generated for each stratum. This information is used when creating the spring model which simulates the soil–pile interaction as seen in Fig. 5b. Finally, to consider the hydraulic added mass, the ocean environment and the sea current were implemented in the software.

Table 1
Material properties of the MPGRTP element.

Part	Volume (m ³)	%	Density (kg/m ³)	Weight (kg)	E-modulus (GPa)	Poisson ratio (dimensionless)
MP	4.83	30.6	7850	37915.5	210	0.3
grout	6.23	39.5	2440	15201.2	60	0.19
TP	4.71	29.9	7850	36973.5	210	0.3
MPGRTP	15.77	100	5712.76	90090.2	165	0.273

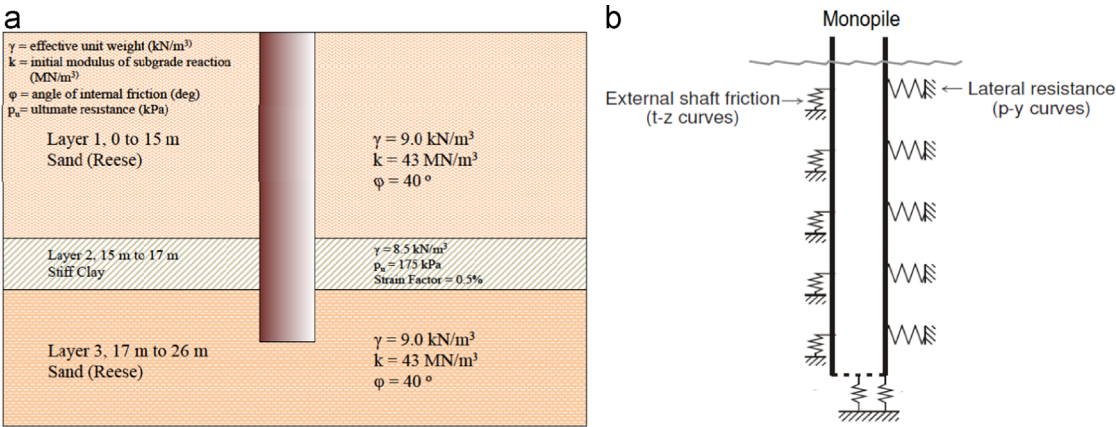


Fig. 5. (a) Soil profile for monopile foundation and (b) spring model of soil–pile interaction.

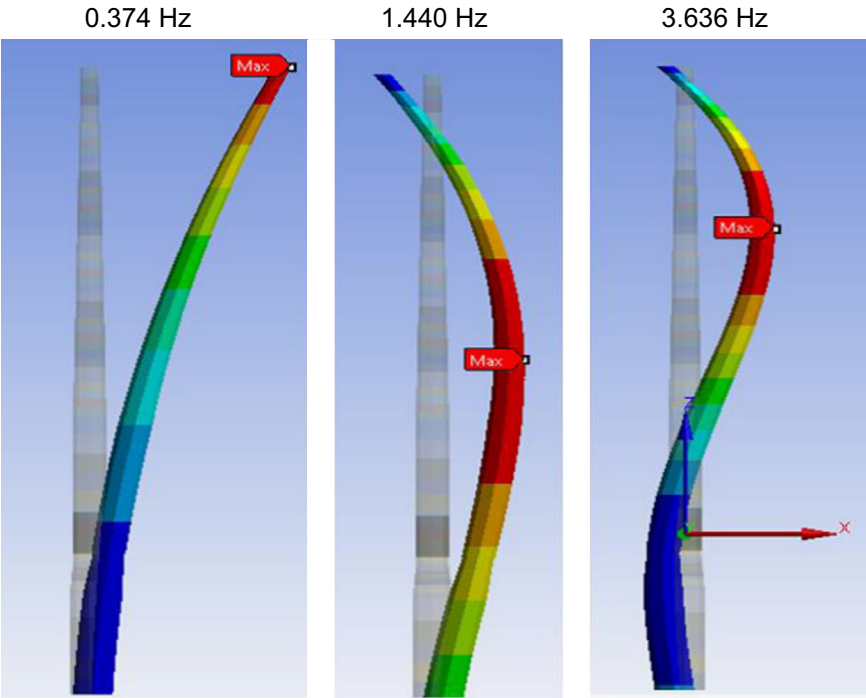


Fig. 6. Fundamental FA tower/foundation modes with their corresponding frequencies obtained through modal analysis in ANSYS.

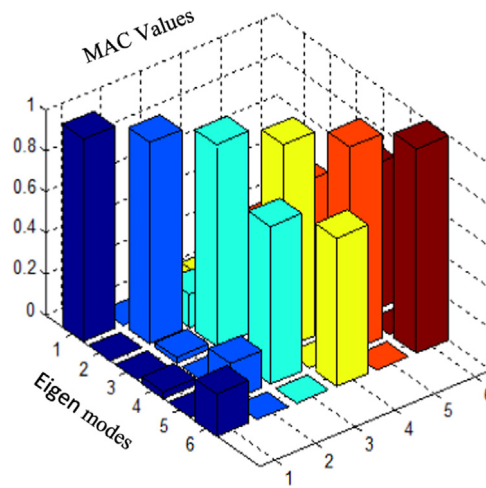


Fig. 7. Graphical illustration of MAC values.

Table 2

MAC values for the 6 fundamental tower/foundation modes.

	FA1	SS1	SS2	FA2	SS3	FA3
FA1	0.987	0	0	0.069	0	0.268
SS1	0	0.981	0.161	0	0.347	0
SS2	0	0.026	0.984	0	0.620	0
FA2	0.032	0	0	0.993	0	0.712
SS3	0	0.181	0.768	0	0.993	0
FA3	0.20	0	0	0.722	0	0.998

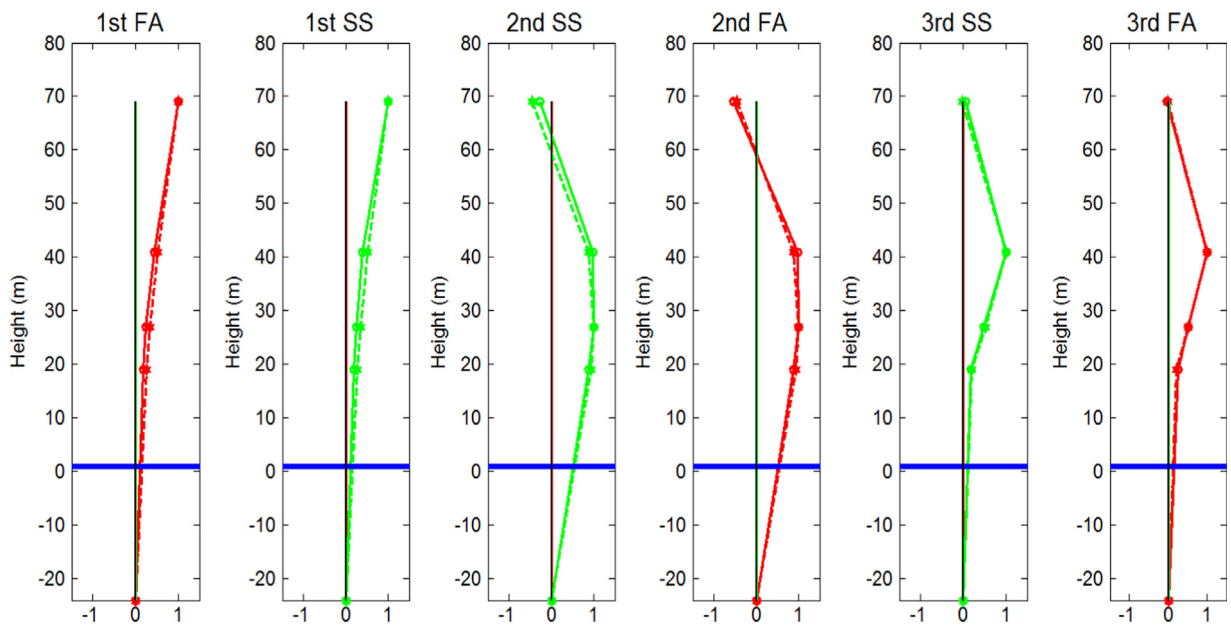


Fig. 8. Overlay plot of the 6 fundamental tower-foundation modeshapes derived from OMA (continuous line with circular points) and FEM (dashed line with star points).

The results of the modal analysis conducted with ANSYS are shown in Fig. 6. In the aforementioned figure only the 3 fundamental modes in the Fore–Aft (FA) direction are presented whereas the Side–Side (SS) modes are omitted as they are identical with the FA due to symmetry in the design of the finite element model. It should be noted that this symmetry

assumption does not hold for the real life structure. For example it is shown in Fig. 3 that the FA2 and SS2 have different frequencies, which differs from the results of the finite element model.

In order to check the accuracy of the numerically obtained mode shapes, a comparison with the experimentally obtained mode shapes is done in terms of the Modal Assurance Criterion (MAC) [23] as follows:

$$MAC(\boldsymbol{\varphi}_{\text{exp},i}, \boldsymbol{\varphi}_{\text{FEM},i}) = \frac{|\boldsymbol{\varphi}_{\text{FEM},i}^T \boldsymbol{\varphi}_{\text{exp},i}|^2}{(\boldsymbol{\varphi}_{\text{exp},i}^T \boldsymbol{\varphi}_{\text{exp},i})(\boldsymbol{\varphi}_{\text{FEM},i}^T \boldsymbol{\varphi}_{\text{FEM},i})} \quad (18)$$

The results depicted in Fig. 7 and their corresponding values summarized in Table 2 indicate the good agreement between the modes obtained from the measurements and the modes obtained from the FEM analysis. The comparison is done at the reference sensor points. A comparative plot of the mode shapes obtained with the OMA identification process and the ones obtained from the finite element model is given in Fig. 8. In Table 3 a comparison between the frequencies obtained from FEM and the OMA identified frequencies is given.

Table 3

Comparison between the FEM and the OMA identified frequencies.

Modes	FEM frequencies (Hz)	OMA frequencies (Hz)	Relative error=(FEM – OMA)/OMA [%]
FA1	0.374	0.361	3.6
SS1	0.374	0.366	2.19
SS2	1.440	1.449	0.62
FA2	1.440	1.560	7.69
SS3	3.636	3.579	1.59
FA3	3.636	3.893	6.6

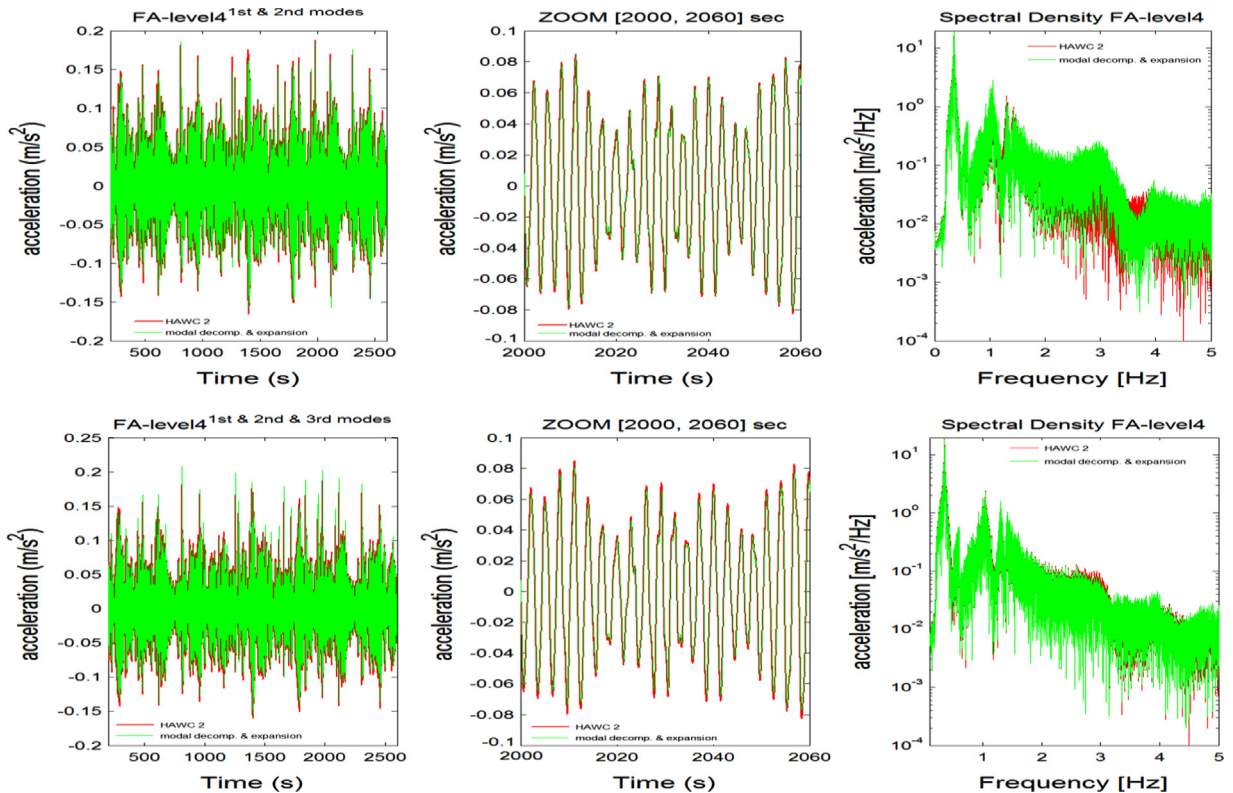


Fig. 9. Time domain and frequency domain comparison of predicted (green) and simulated accelerations (red) at the lowest sensor level for parked conditions in FA direction using 2 modes and 2 sensors (top) or 3 modes and 3 sensors (bottom). The wind speed is 5. m/s (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

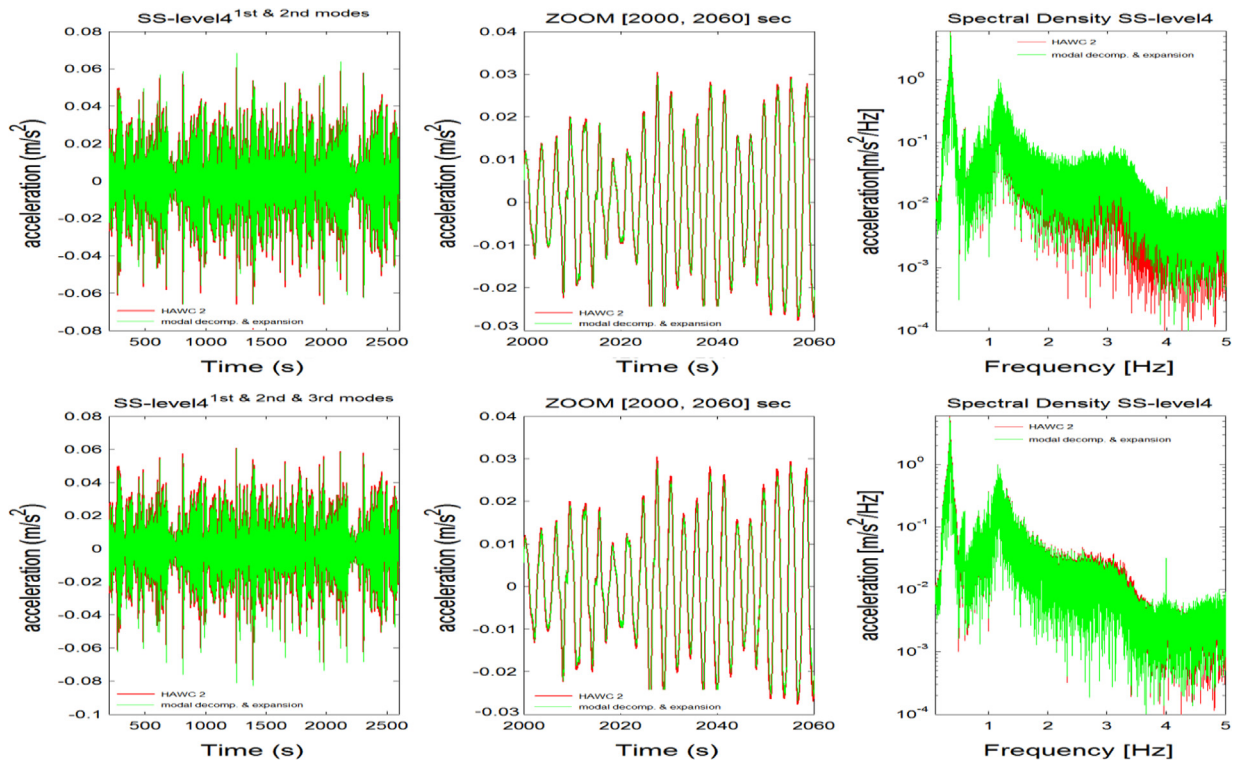


Fig. 10. Time domain and frequency domain comparison of predicted (green) and simulated accelerations (red) at the lowest sensor level for parked conditions in SS direction using 2 modes and 2 sensors (top) or 3 modes and 3 sensors (bottom). The wind speed is 5 m/s (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

Based on the presented comparison results between the OMA and FEM output concerning the modal parameters of the turbine, it was decided that no further tuning of the finite element model is required at this stage. Obviously a more rigorous updating would be possible; this was however considered to be outside the scope of this paper.

6. Response estimation

6.1. Virtual simulated conditions using HAWC2

All the simulations are carried out for 50 min and the first 10 min segment is removed to avoid the transient behavior at the beginning of the simulations. The 40-min simulations are performed with a sampling frequency of 50 Hz giving a total of 120,000 data points. The virtual time domain acceleration responses at the installed sensor locations (69 m, 41 m, 27 m and 19 m above LAT) are obtained and used for the analysis. Both parked and rotating conditions are evaluated. In order to evaluate the robustness of the method, both conditions were evaluated at different wind speeds. The wind speeds examined in parked conditions are 5, 10 and 18 m/s respectively and for a rotating turbine the wind velocities were chosen to be 5, 15 and 25 m/s respectively. Finally, an overspeed stop test was also examined.

6.1.1. Parked and rotating simulated conditions

At a first stage a comparative study between the results obtained when using two modes and two sensor levels or three modes and three sensor levels is presented for both parked and rotating conditions in the Fore–Aft and Side–Side direction. The comparison is done in order to highlight the importance on making proper decisions on the number of modes and sensor levels necessary to achieve optimal prediction quality. The wind speed of 5 m/s is chosen as a basis for comparison of both parked and rotating conditions. At a next stage a quantification of the quality of predictions for all simulated wind speeds is given.

In the next figures, the results of the FA and SS acceleration prediction at the lowest sensor level (level 4, $Z = +19$ m above LAT) are presented corresponding both to the parked and rotating conditions of the wind turbine. Red color corresponds to acceleration derived from the virtual accelerometer of the HAWC2 simulations whereas the predicted accelerations are shown in green.

From the graphs in Figs. 9–12 it is made clear that there is a good agreement between the virtual data derived from HAWC simulations and the predicted data derived from the proposed modal decomposition and expansion technique. The

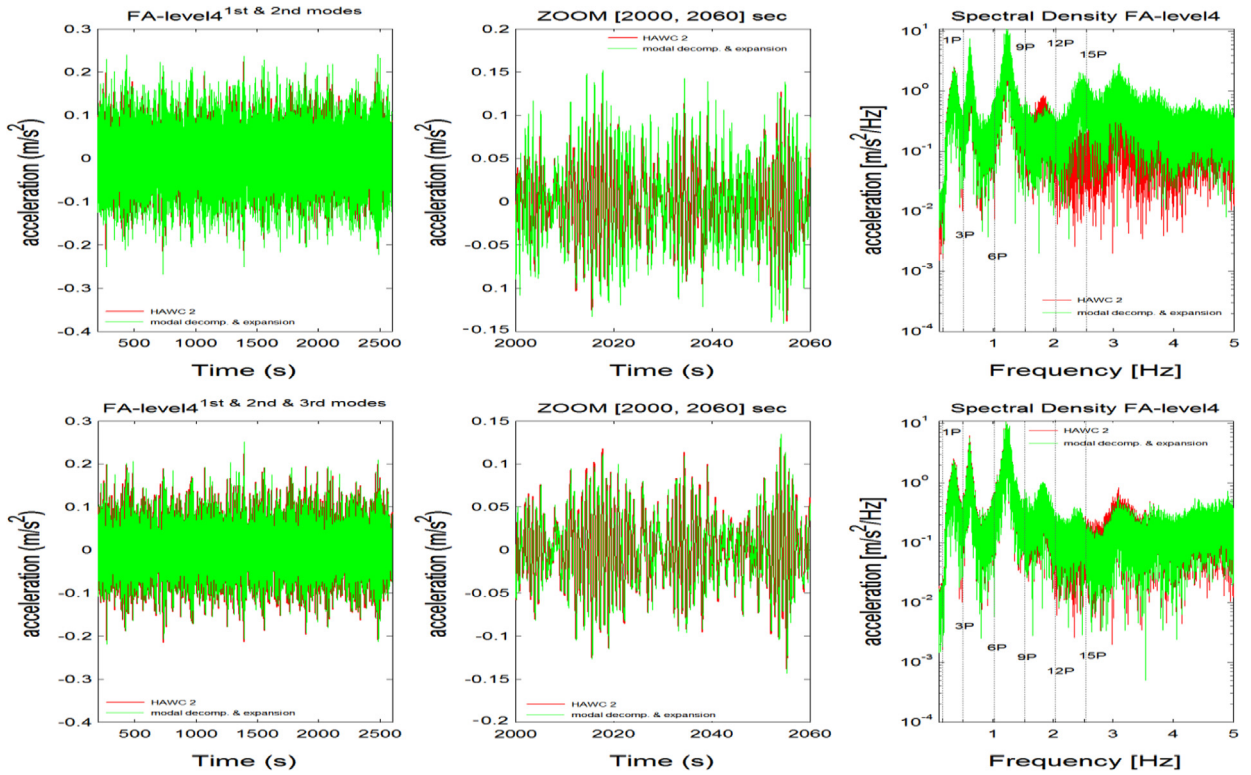


Fig. 11. Time domain and frequency domain comparison of predicted (green) and simulated accelerations (red) at the lowest sensor level for rotating conditions in FA direction using 2 modes and 2 sensors (top) or 3 modes and 3 sensors (bottom). The wind speed is 5 m/s. The nominal rotation frequency 1 P at 0.17 Hz (10 rpm) and the higher order harmonics up to 15P are shown in the frequency domain plot (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

agreement is good both in terms of amplitude and in terms of temporal evolution in the time domain as well as in the frequency domain for both parked and rotating conditions of the wind turbine both in the FA and SS direction.

In order to visualize to which extent and at which frequency bands each mode is dominantly contributing to the total vibrational response one can plot the Fourier spectra of the modal coordinates, Fig. 13. It can nicely be observed, that in the frequency bands around the resonance frequencies the corresponding mode has the most dominant contribution, whereas in the other frequency bands the dynamic behavior is described by a linear combination of several acting modes.

Fig. 14 gives an overview of the TRAC and FRAC values for the different cases and the different states of the wind turbine in FA and SS direction. The FRAC has been calculated for different frequency bands of interest.

In parked conditions, the dynamic behavior can almost be completely described by the first two modes. This statement is true not only for the lowest frequency range where there is an outperformance, but also for the largest frequency bands in which the FRAC and TRAC values calculated are sufficiently high (above 0.994). As the frequency band of interest increases, the use of 3 modes and 3 sensor levels implies a slight improvement on the overall prediction quality.

In rotating conditions, it is noticeable that the implementation of a 3rd mode and a 3rd sensor level in the prediction process significantly increases the FRAC values, indicating that this kind of determined system (3 sensors–3 modes) is imperative for more precise and accurate prediction of the acceleration response. In the case of using 2 modes and 2 sensor levels in the frequency band of 0.05–5 Hz, the FRAC drops below 0.9. This behavior is especially observed in large frequency bands, whereas in the lowest frequency band a secure and sufficient prediction can be made by using only the first two modes.

It should be emphasized that the FRAC value coincides with the TRAC value when examined in the full frequency band ([0, 50 Hz]) but differs in other frequency bands of interest allowing the FRAC to be used for the interpretation of the quality of predictions at specific frequency ranges.

In the figure that follows (Fig. 15), the quality indicators described in Section 2.4 are plotted for all the different simulated wind speeds in both parked and rotating conditions in FA direction. The results in the SS direction are omitted since they exhibit similar behavior. The frequency band that ranges from 0.05 Hz to 5 Hz is chosen for plotting the FRAC indexes, since it is the most representative and able to capture the most dominant modes acting at a certain direction (FA or SS).

For all the simulated wind speeds in parked conditions of the wind turbine, the TRAC, FRAC and mean absolute error in the frequency domain do not significantly change if two sensor levels and two modes or three sensor levels and three modes are used in the analysis. Moreover, the correlation between the simulated and predicted signals is sufficiently high if a

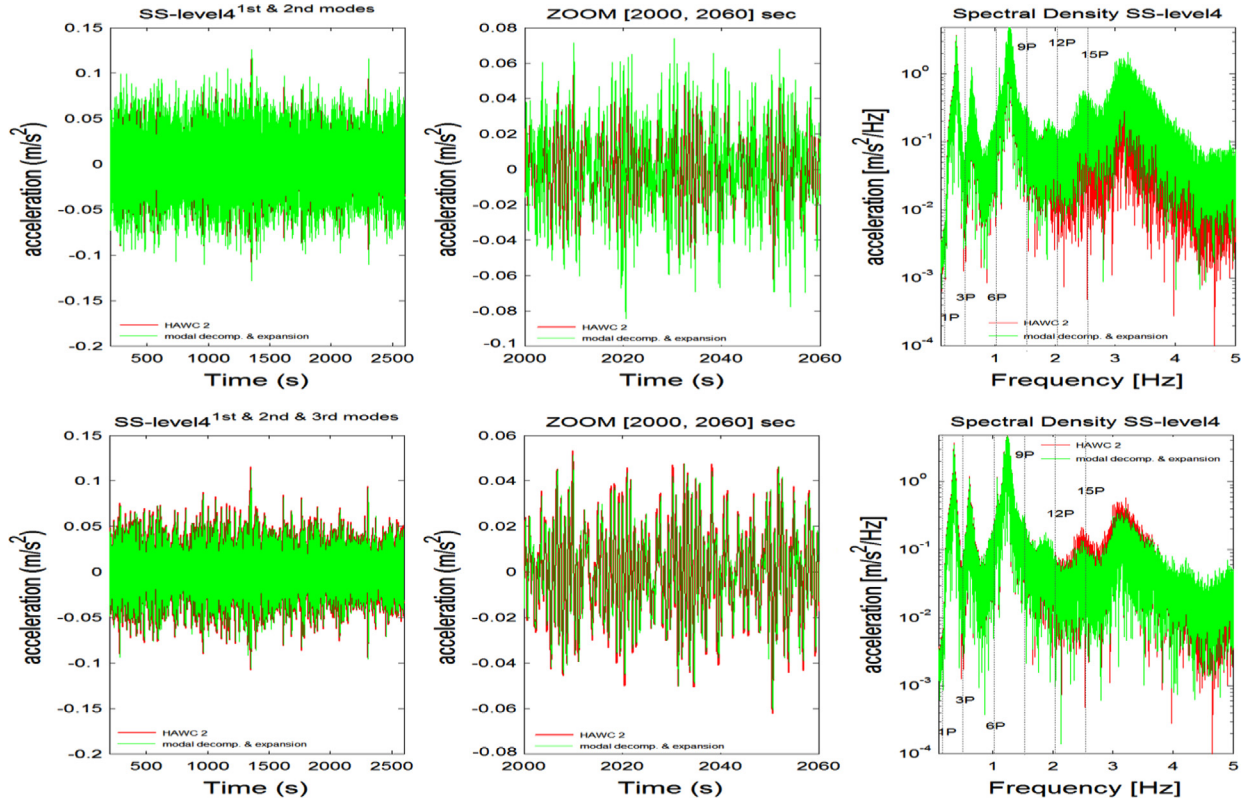


Fig. 12. Time domain and frequency domain comparison of predicted (green) and simulated accelerations (red) at the lowest sensor level for rotating conditions in SS direction using 2 modes and 2 sensors (top) or 3 modes and 3 sensors (bottom). The wind speed is 5 m/s. The nominal rotation frequency 1 P at 0.17 Hz (10 rpm) and the higher order harmonics up to 15 P are shown in the frequency domain plot (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

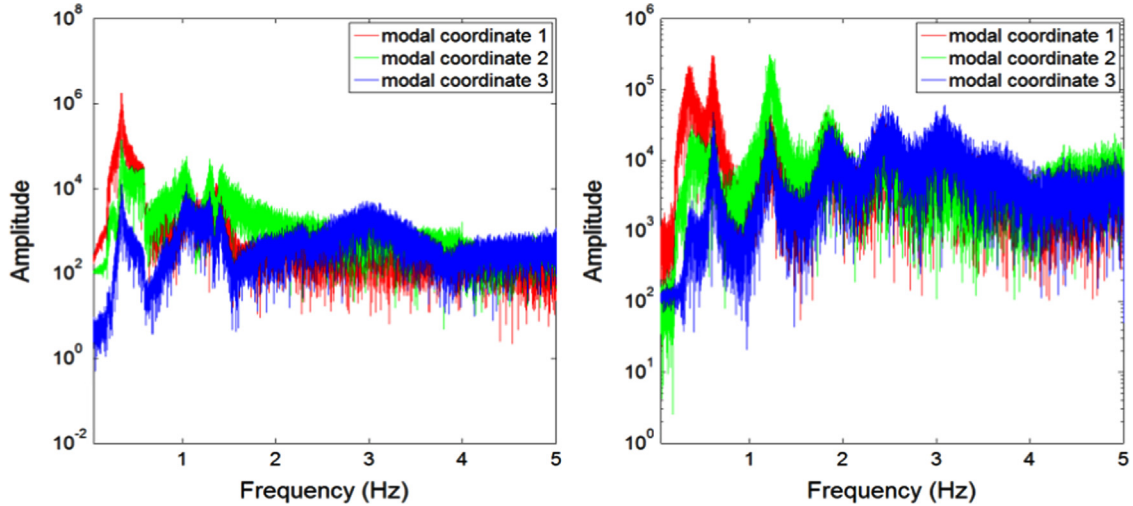


Fig. 13. Fourier spectrum of the modal coordinates for parked and rotating conditions in Fore-Aft (FA) direction for the wind speed of 5 m/s.

determined system consisting of 2 sensor levels and 2 modes is taken into account, indicating that the first two modes are sufficient enough to describe the dynamic behavior of the parked wind turbine.

When the turbine is under normal rotating conditions, the contribution of the third mode and the information obtained from a third accelerometer level is providing much more accurate results, indicated by the significant increase in the TRAC and FRAC indexes as well as a significant decrease in the mean absolute error. At low wind speed (5 m/s) where the wind turbine is almost idling, two modes and two sensor levels can already achieve a high-level prediction.

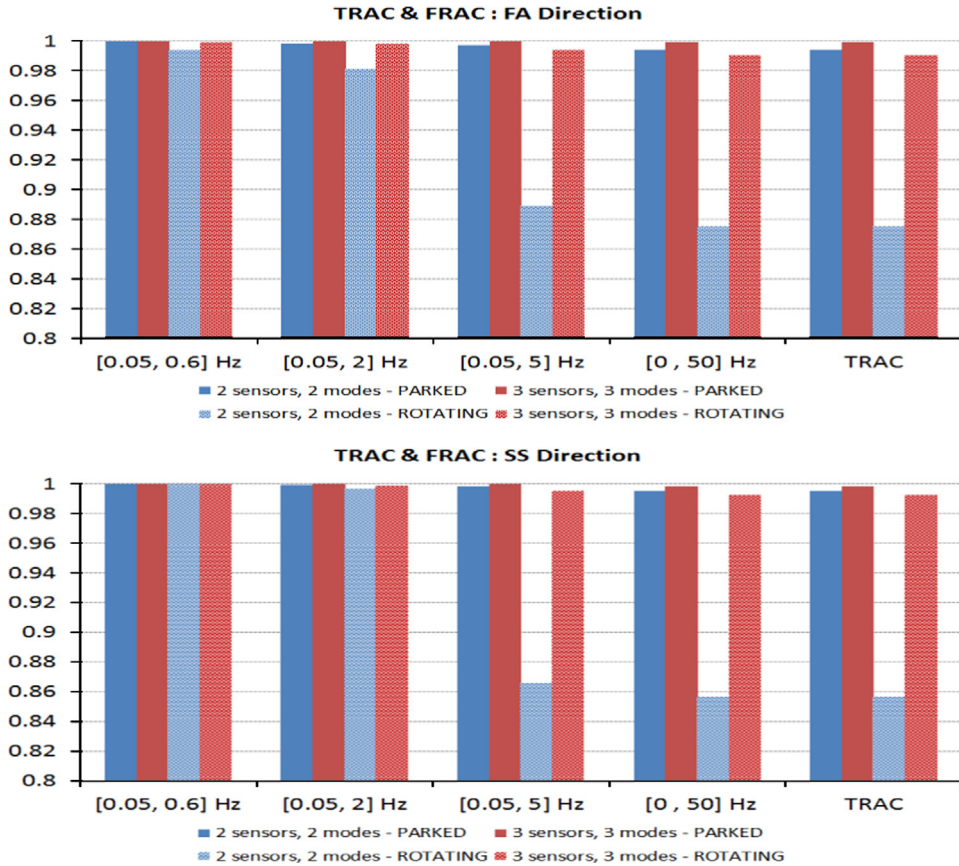


Fig. 14. TRAC value and Frequency Response Assurance Criterion (FRAC) calculated in frequency bands of interest using either 2 modes and 2 sensors (blue) or 3 modes and 3 sensors (red). Results for both parked (solid bars) and rotating conditions (hatched bars) in both FA and SS direction at the common wind speed of 5 m/s (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

6.1.2. Overspeed stop test

In this section, an overspeed stop test is analyzed. This type of event is of utmost importance since it can severely contribute to the overall fatigue damage of the structure and can be crucial to quantify the reliability and the lifetime of offshore wind turbines both in the design phase as well as during its life-cycle.

In the analyzed overspeed stop, the wind speed is increasing from 4 m/s to 6.5 m/s and at time 85 s the pitch angle changes quickly from 0 to 88.2°. The thrust due to sudden collective pitch angle variation excites the tower mainly in the wind direction (FA mode). A 5-min simulation with a sampling frequency of 12.5 Hz is run, giving a total of 3750 data points. The contribution of only two modes in the FA direction with the acceleration data obtained from the highest two sensor levels are considered in order to predict the dynamic response at the sensor level 3, situated 27 m above the LAT. As described in Section 6.1.1, red color is used for the virtual (simulated) accelerations obtained from HAWC2 aeroelastic simulation tool, whereas the acceleration estimated with the methodology proposed in this paper are shown in green.

The simulated and the predicted acceleration signals match perfectly both in the time domain and in the frequency domain, Fig. 16. The results indicate a good agreement both in terms of amplitude and in terms of temporal evolution. It should be emphasized that the use of the first two modes is capable of predicting the dynamic behavior of the wind turbine with high precision even in such kind of extreme loading condition. As summarized in Table 4, the good correlation between the simulated and the predicted acceleration time domain signals results in a high TRAC value and a low mean absolute error. This can be explained due to the fact that the overspeed stop mainly excites the first mode. Similarly, the frequency domain comparison of the signals exhibit high values of FRAC in various examined frequency bands (Table 4).

6.2. Real-life loading conditions of the wind turbine

In the previous subsection, the validation of the proposed modal decomposition and expansion technique was based on different virtual wind-wave loading conditions of the wind turbine by making use of simulated responses. In this subsection, the use of real data obtained from a monitoring campaign on an offshore Vestas V90 3 MW wind turbine on a monopile foundation will be used for the analysis.

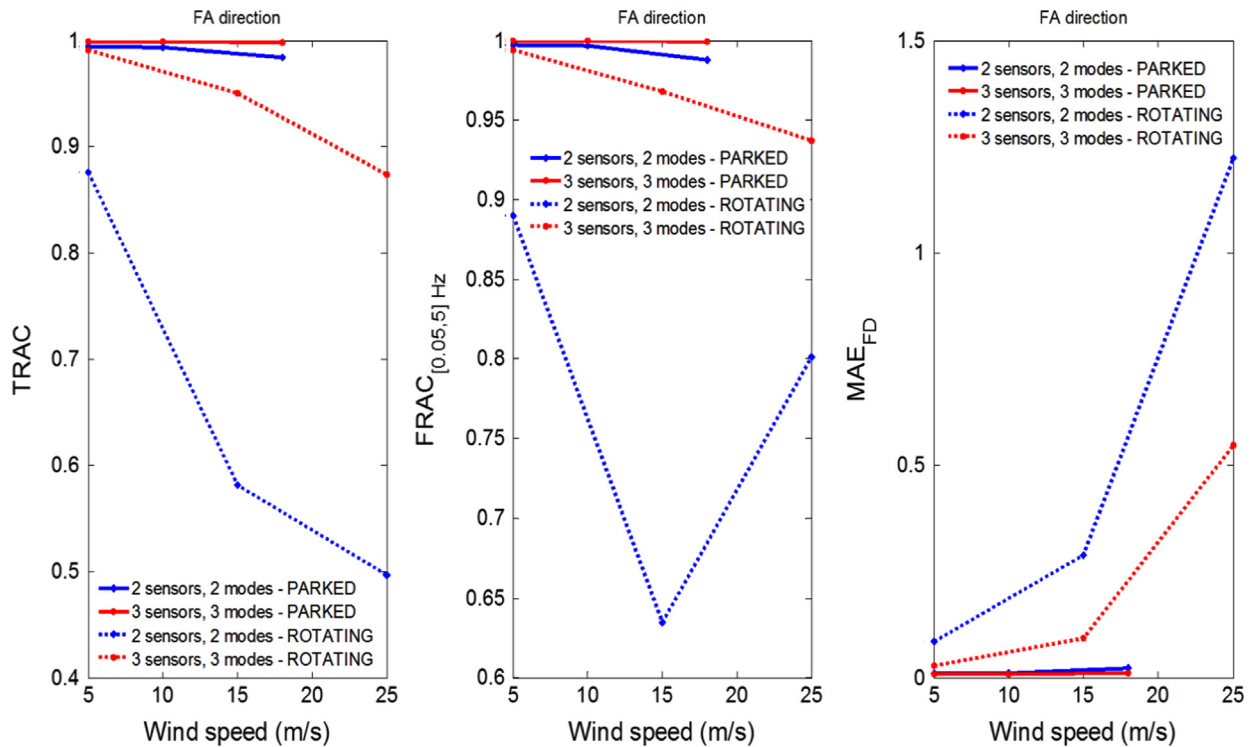


Fig. 15. Indicative quality indicators (TRAC, FRAC, MAE_{FD}) as a function of simulated wind speeds for the FA direction of the parked (continuous lines) and rotating conditions (dashed lines) of the wind turbine.

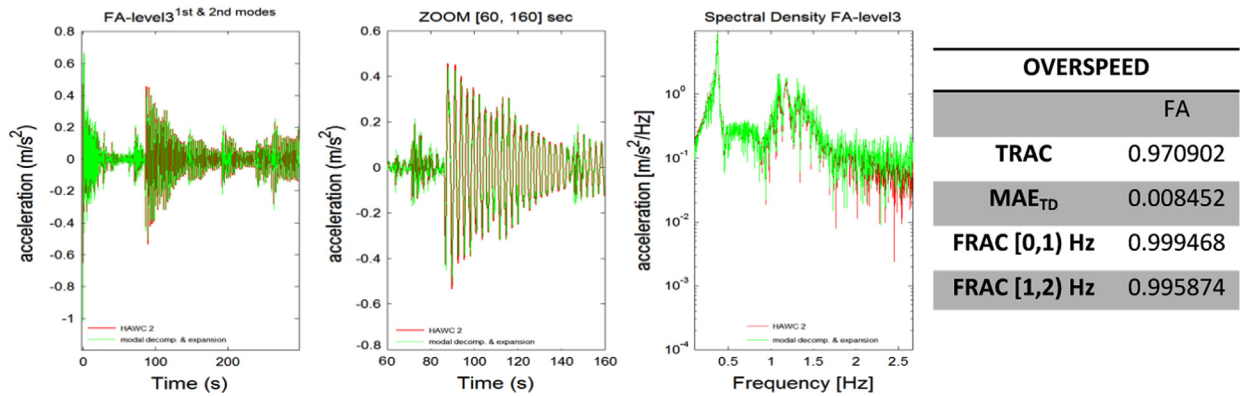


Fig. 16. Time domain and frequency domain comparison of predicted (green) and simulated accelerations (red) at the third sensor level (Z=27 m above LAT) in FA direction for the overspeed stop test. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 4

Summarizing table with values of quality indicators utilized to quantify the prediction.

OVERSPEED	
TRAC	FA 0.970902
MAE _{TD}	0.008452
FRAC [0,1] (Hz)	0.999468
FRAC [1,2] (Hz)	0.995874

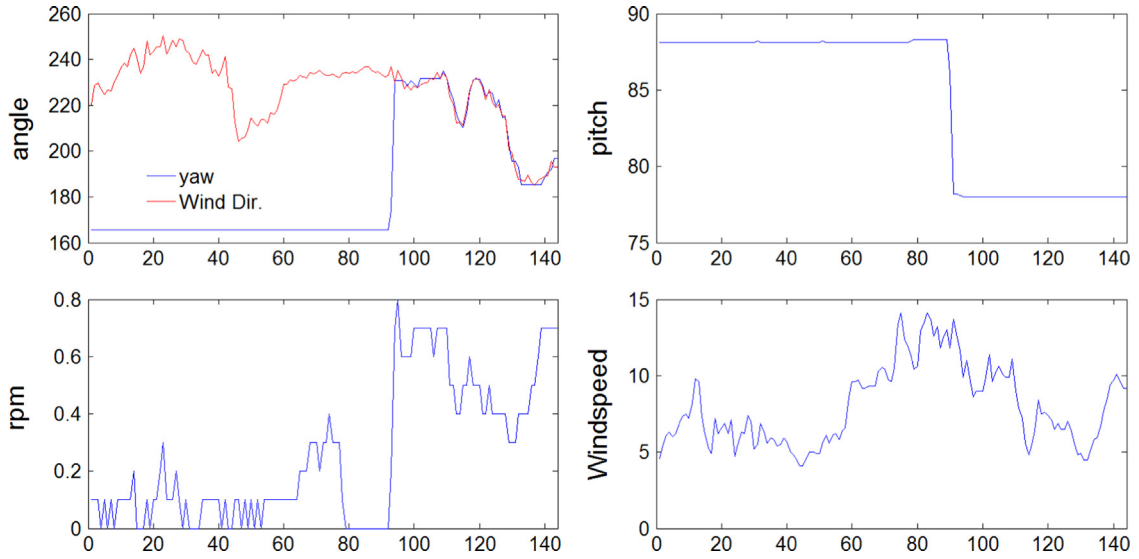


Fig. 17. Full day measured SCADA and meteorological data for parked conditions of the wind turbine.

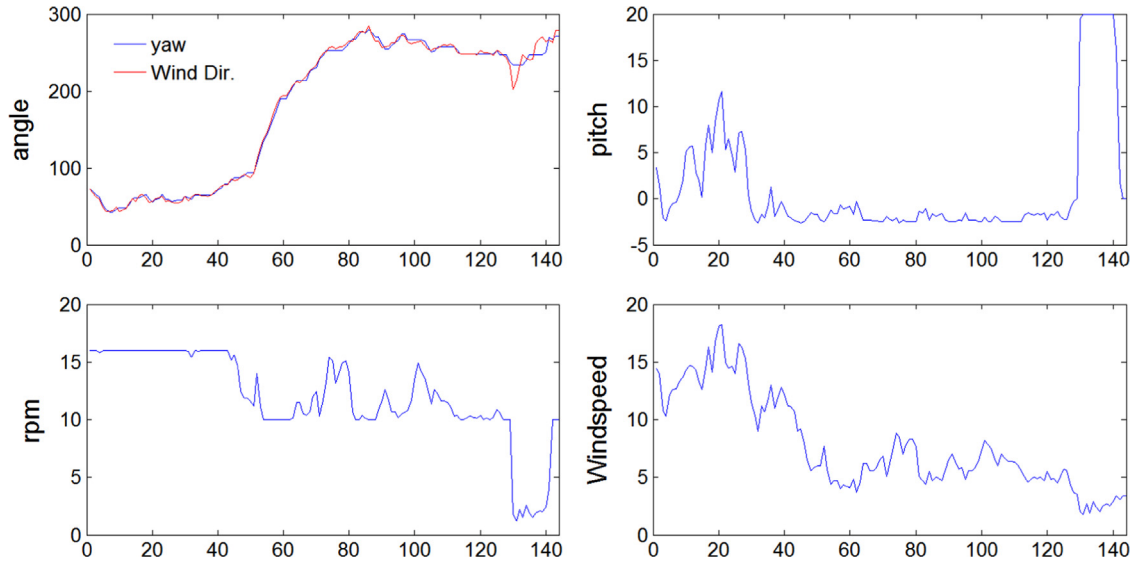


Fig. 18. Example of full day measured SCADA and meteorological data for rotating conditions of the wind turbine.

In order to monitor varying and environmental operating conditions of the wind turbine during the measurements, SCADA data (power, rotor speed, pitch angle, and nacelle direction) and ambient data (wind speed, wind direction, significant wave height, air temperature, etc.) are being collected at 10 min intervals. A full day of parked and a full day of rotating conditions are analyzed. A total of 144 ten-minute files, each of them provided with a sampling frequency of 20 Hz and 120,00 data points, were analyzed in order to obtain the predicted accelerations both in the FA and in the SS direction. The SCADA and meteorological data corresponding to the full day of parked and the full day of rotating conditions of the wind turbine are given in Figs. 17 and 18.

6.2.1. Parked conditions

An indicative plot of the time and frequency domain comparison between the measured and predicted acceleration signals in parked conditions in the FA and SS direction is presented in Fig. 19. In this figure, it should be highlighted that three modes and three sensors in each direction (FA or SS) are considered. The results of the prediction of response accelerations for the full day of parked or idling state of the wind turbine are shown in Fig. 20 for the FA direction and in Fig. 21 for the SS direction.

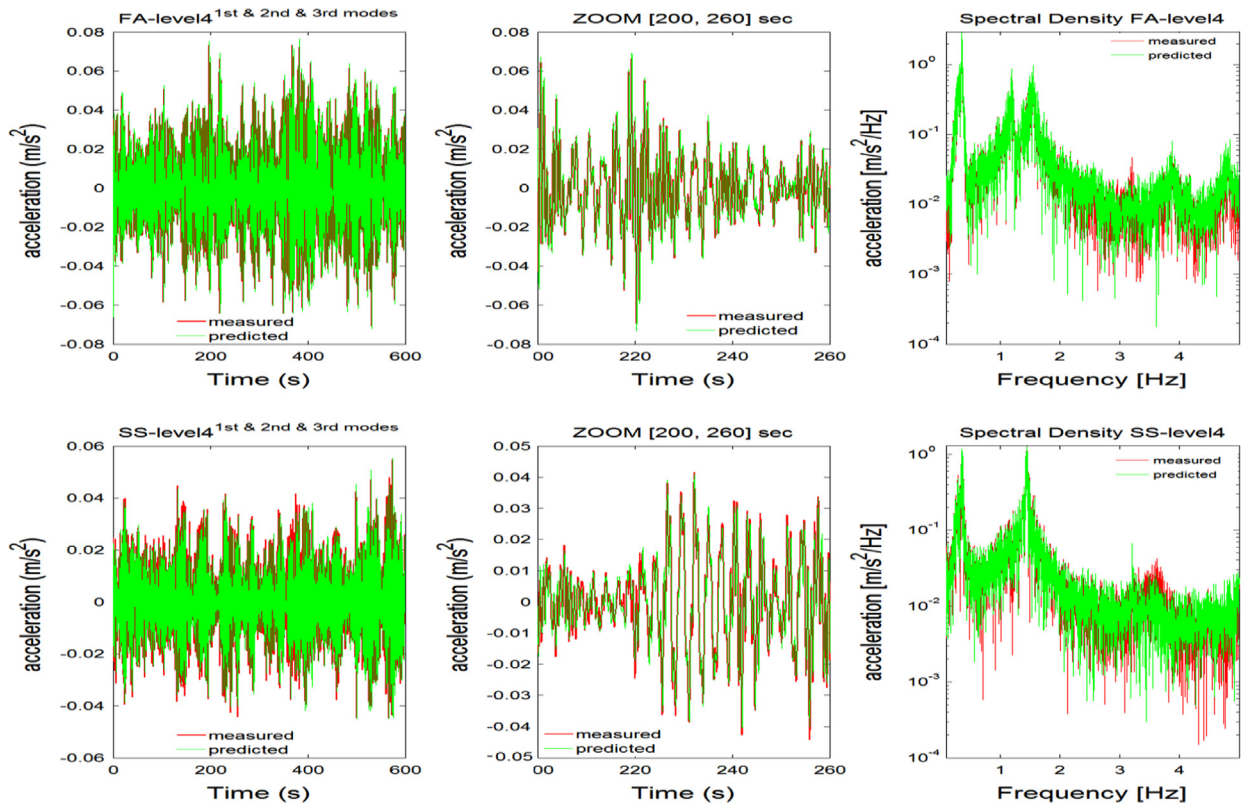


Fig. 19. Time domain and frequency domain comparison of predicted (green) and measured accelerations (red) at the lowest sensor level in FA and SS direction for parked conditions of the wind turbine (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

In general it can be said that the quality of prediction is very good. In most cases the TRAC and the FRAC are higher than 0.9 when considering 3 modes and higher than 0.8 when considering 2 modes in FA direction or higher than 0.7 when considering 2 modes in SS direction. During the hours of the day where the wind is constantly blowing at relative high wind speeds (above 9 m/s – see indexes 60:110 in Fig. 17) there is clearly observed a decrease in TRAC and FRAC values or increase in MAE when using a 2 sensor–2 mode determined system. At the same region, though, the system that takes into consideration 3 modes and 3 sensor levels is capable of fully describing the dynamic behavior of the wind turbine, thus leading to high values of the quality indicators with reduced errors. Specific changes in the state of the wind turbine e.g. from idling to parked (see rpm indexes 79:95 in Fig. 17) and the ambient conditions might give rise to measurement errors that might lead to lower values of the quality indicators. It is part of future work to understand the specific reasons why in certain conditions the prediction is less accurate.

6.2.2. Rotating conditions

Same amount of data as described in Section 6.2.1 is collected and post-processed for a measurement day where the wind turbine is under normal operating conditions. The results are summarized in the following figures (Figs. 22–24)

As it is made clear from the figures above, the implementation of a third sensor level and a third mode contributing to the prediction process, significantly improves the prediction quality. It can be concluded that in operating conditions the use of only 2 modes and 2 sensor levels is not enough to fully describe the dynamic behavior of the wind turbine, resulting in TRAC and FRAC values below 0.8. This behavior is observed both in FA and SS direction for a whole day of rotating state of the wind turbine. The TRAC and FRAC indexes, for the case of using 3 modes, are almost approaching the unity for the FA direction and high values around 0.9 for the SS direction, proving that there is a good match between the measured and predicted time domain and frequency domain acceleration signals (Figs. 23 and 24), leading to low values of mean absolute errors.

A sudden change in the pitch angle from 0° to 20° and back, as indicated in Fig. 18 (see indexes 138:144), causes a fluctuation in the obtained TRAC and FRAC values both in the FA and SS direction [28]. Similar to the parked conditions, for fluctuations in the wind speed above 9 m/s (see indexes 0:32 in Fig. 18) there is an observed decrease in TRAC and FRAC values or increase in MAE when using a 2sensor–2mode determined system. At the same region, though, the system that takes into consideration 3 modes and 3 sensors is capable of fully describing the dynamic behavior of the wind turbine, thus leading to high values of the quality indicators with reduced errors both in the time domain and in the frequency domain, both in FA and SS direction.

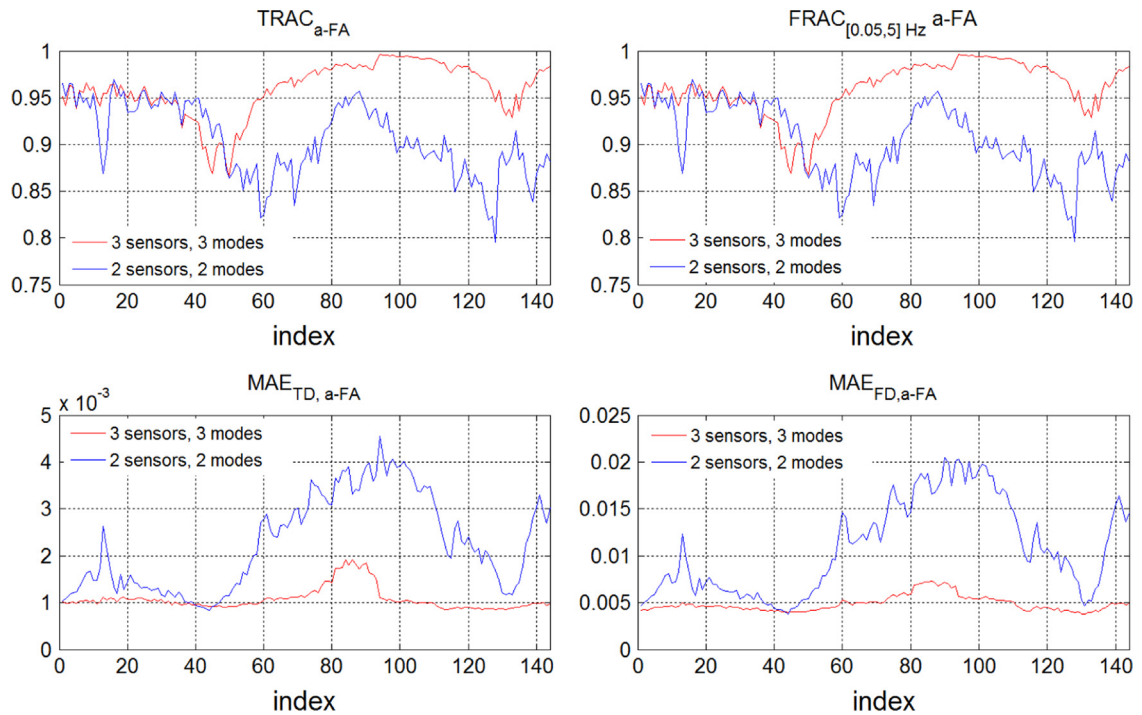


Fig. 20. TRAC, FRAC and mean absolute error indexes as tools to quantify the quality of acceleration predictions at the lowest sensor level for a full day of parked conditions of the wind turbine. Results in FA direction notifying the impact of using a 2sensor–2mode or 3sensor–3mode determined system to predict the vibrational response at the lowest sensor level.

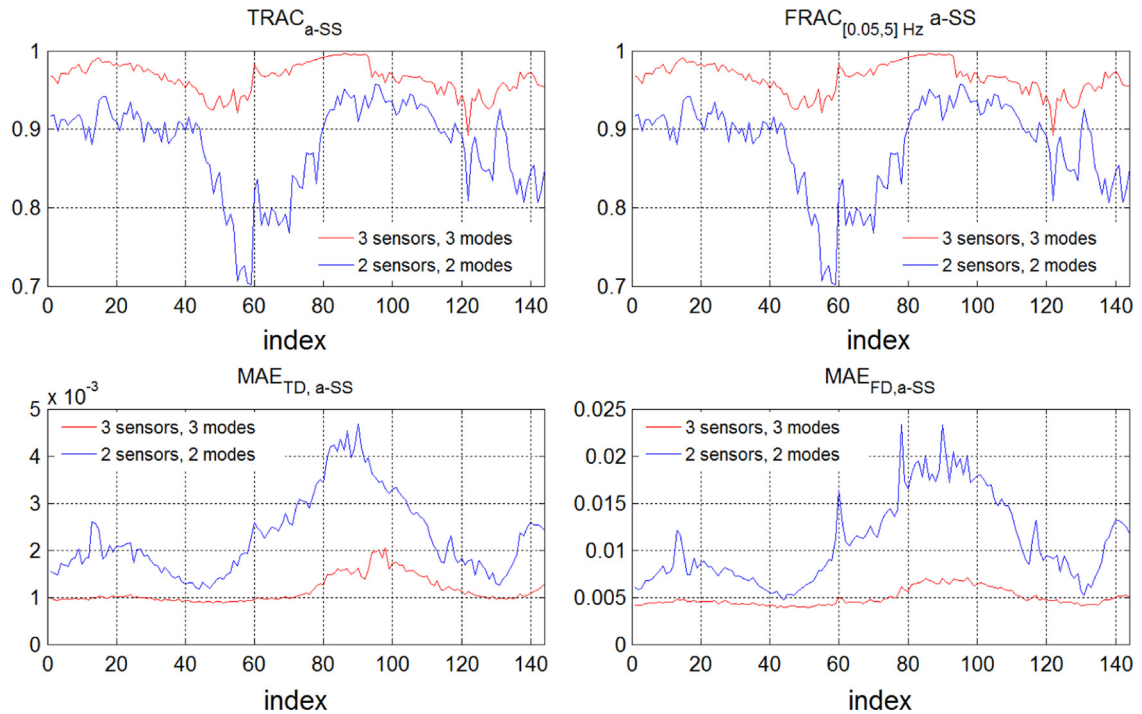


Fig. 21. TRAC, FRAC and mean absolute error indexes as tools to quantify the quality of acceleration predictions at the lowest sensor level for a full day of parked conditions of the wind turbine. Results in SS direction notifying the impact of using a 2sensor–2mode or 3sensor–3mode determined system to predict the vibrational response at the lowest sensor level.

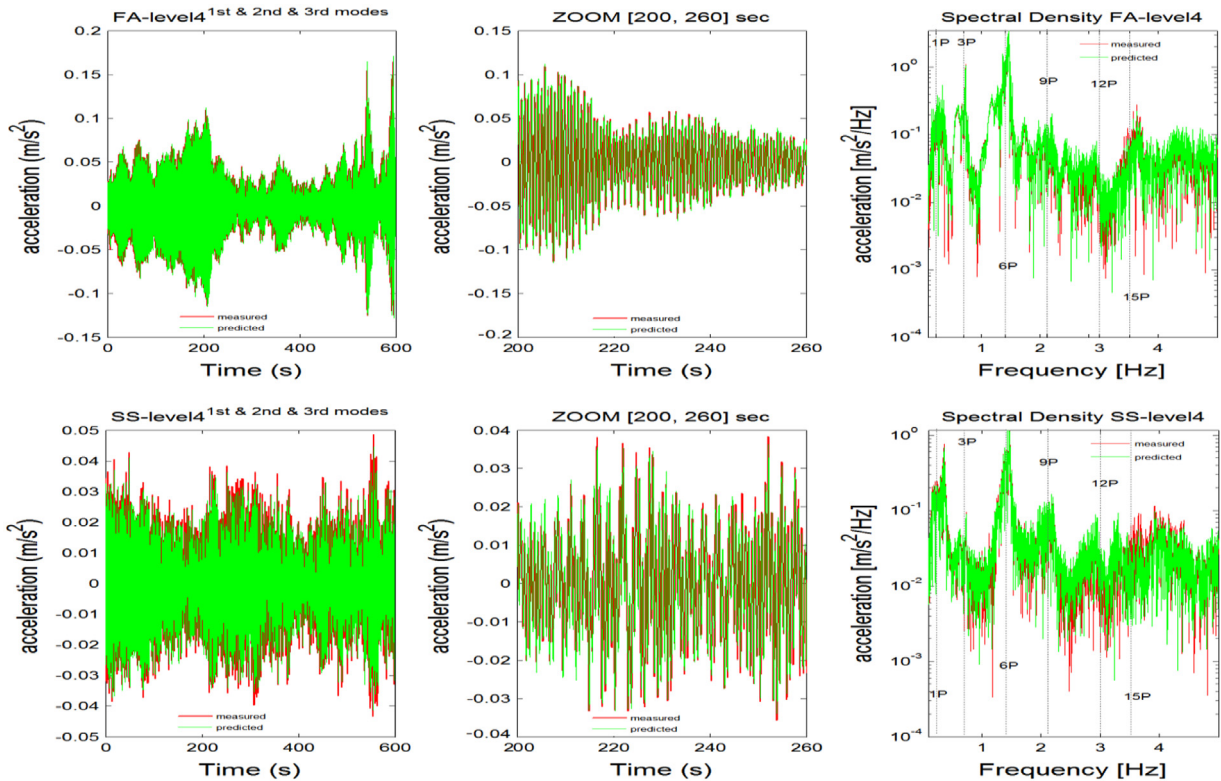


Fig. 22. Time domain and frequency domain comparison of predicted (green) and measured accelerations (red) at the lowest sensor level in FA and SS direction for rotating conditions of the wind turbine. The nominal rotation frequency 1 P at 0.235 Hz (14.1 rpm) and the higher order harmonics up to 15 P are shown in the frequency domain plot for the indicative timestamp (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

6.2.3. Overspeed stop real-life test

In this paragraph, an overspeed stop test is considered and acceleration data from the installed accelerometers are collected. During such an overspeed test the wind speed initially is the minimum required (6.5 m/s). Then the wind turbine speeds up until it reaches 19.8 rpm. At that moment the wind turbine is automatically shut down. During this stop, the pitch angle is changed from -2.5 to 88.2° in a couple of seconds. The thrust release due to this sudden collective pitch variation excites the tower mainly in the wind direction (FA). The contribution of the first two modes in FA direction together with the corresponding acceleration information obtained from two sensors installed in the upper levels of the tower are considered. Red color corresponds to acceleration derived from the measured accelerations derived from accelerometers mounted on the structure whereas the acceleration estimated with the modal decomposition and expansion methodology proposed in this paper are shown in green.

As shown in the above figure (Fig. 25), the modal decomposition and expansion algorithm is capable of providing predictions of high accuracy and precision even though the loading condition examined in this case is extreme.

7. Conclusions and future work

This paper presents the results of a dynamic response prediction method applied on an offshore wind turbine on a monopile foundation. The proposed method is based on a modal decomposition and expansion approach that is used to estimate the response at unmeasured locations by combining a limited set of acceleration response measurements and a finite element (FE) model. The method has proven to be fast, easy to implement, reliable and effective for the prediction of acceleration signals. It provides time domain and frequency domain predicted acceleration signals in good agreement with both simulated and real measured acceleration signals both in terms of amplitude and in terms of temporal evolution. The method has been examined for several cases corresponding to different operational states of the wind turbine (parked/idling state, rotating state and overspeed stop extreme case). For all cases the method was able to generate good results. The small differences that could be found in certain cases can be attributed to changing operational and/or ambient conditions. This result was expected as changing operational and ambient conditions can affect the dynamic behavior of the wind turbine. Especially changes in the mode shapes will affect the prediction results. Therefore it is considered as part of future work to use case dependent mode shapes by tuning the finite element model for each specific operating condition.

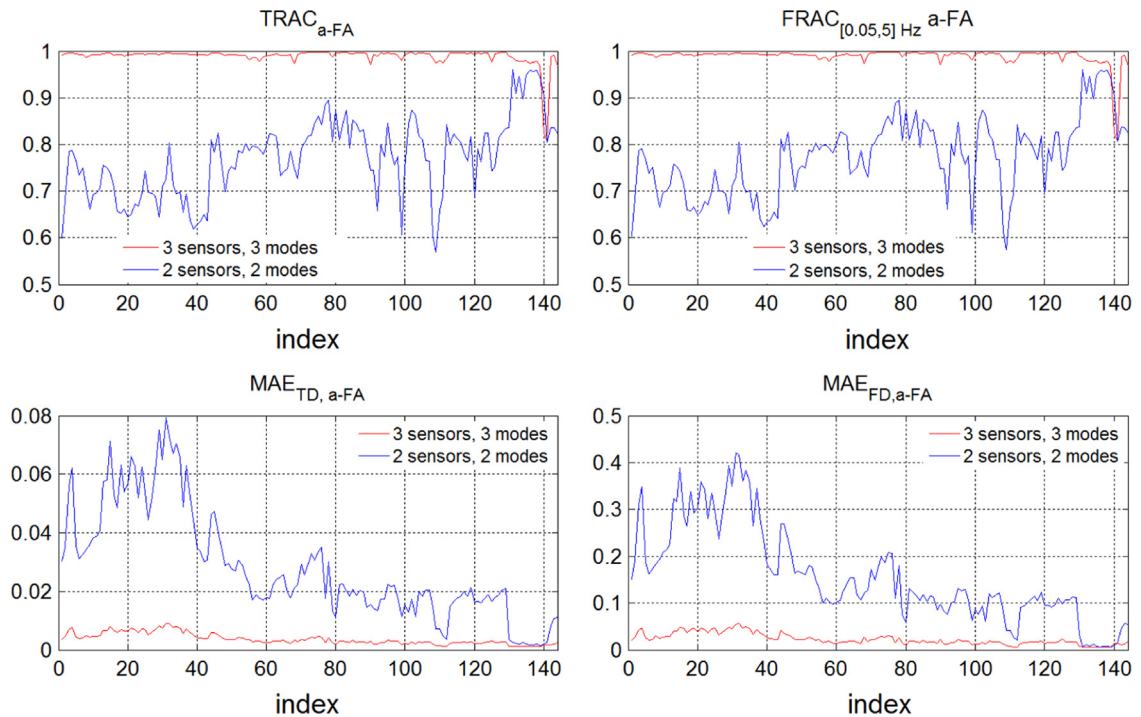


Fig. 23. TRAC, FRAC and mean absolute error indexes as tools to quantify the quality of acceleration predictions at the lowest sensor level for a full day of rotating conditions of the wind turbine. Results in FA direction notifying the impact of using a 2sensor–2mode or 3sensor–3mode determined system to predict the vibrational response at the lowest sensor level.

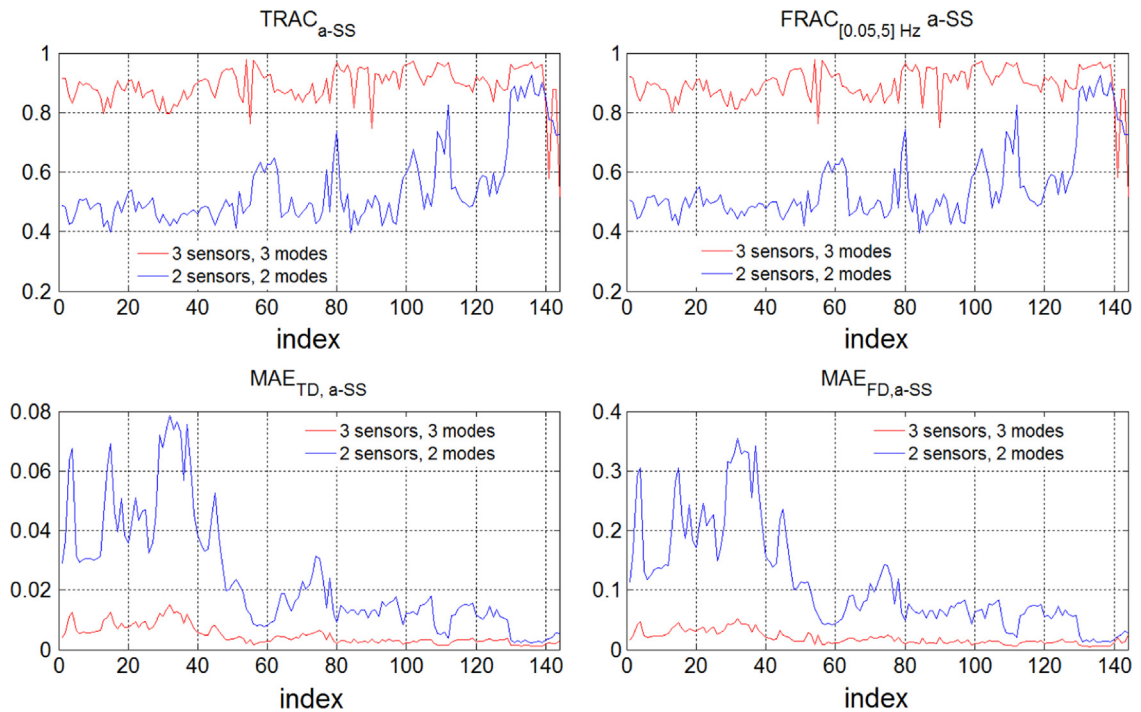


Fig. 24. TRAC, FRAC and mean absolute error indexes as tools to quantify the quality of acceleration predictions at the lowest sensor level for a full day of rotating conditions of the wind turbine. Results in SS direction notifying the impact of using a 2sensor–2mode or 3sensor–3mode determined system to predict the vibrational response at the lowest sensor level.

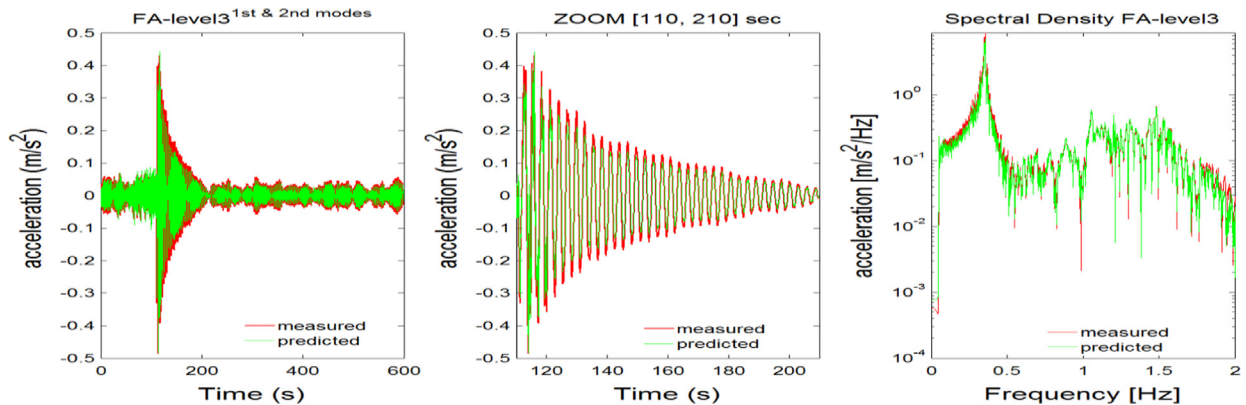


Fig. 25. Time domain and frequency domain comparison of predicted (green) and measured accelerations (red) at sensor level 3 ($Z = +27$ m above LAT) in FA direction for overspeed stop test case (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

Furthermore the current approach is restricted to the prediction of acceleration responses, future research will try to extend this to the prediction of dynamic strain responses that will be validated based on real time strain data obtained from a monitoring campaign on the examined wind turbine.

Acknowledgments

This research has been performed in the frame-work of the Offshore Wind Infrastructure Project (<http://www.owi-lab.be>) and the IWT SBO Project, OptiWind (Grant no: IWT/120029 - OPTIWIND) and the IWT O&O Parkwind project. The authors therefore acknowledge the financial support from The Agency for innovation by Science and Technology (IWT). The authors also gratefully thank the people of Belwind NV for their continuous support within this project.

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